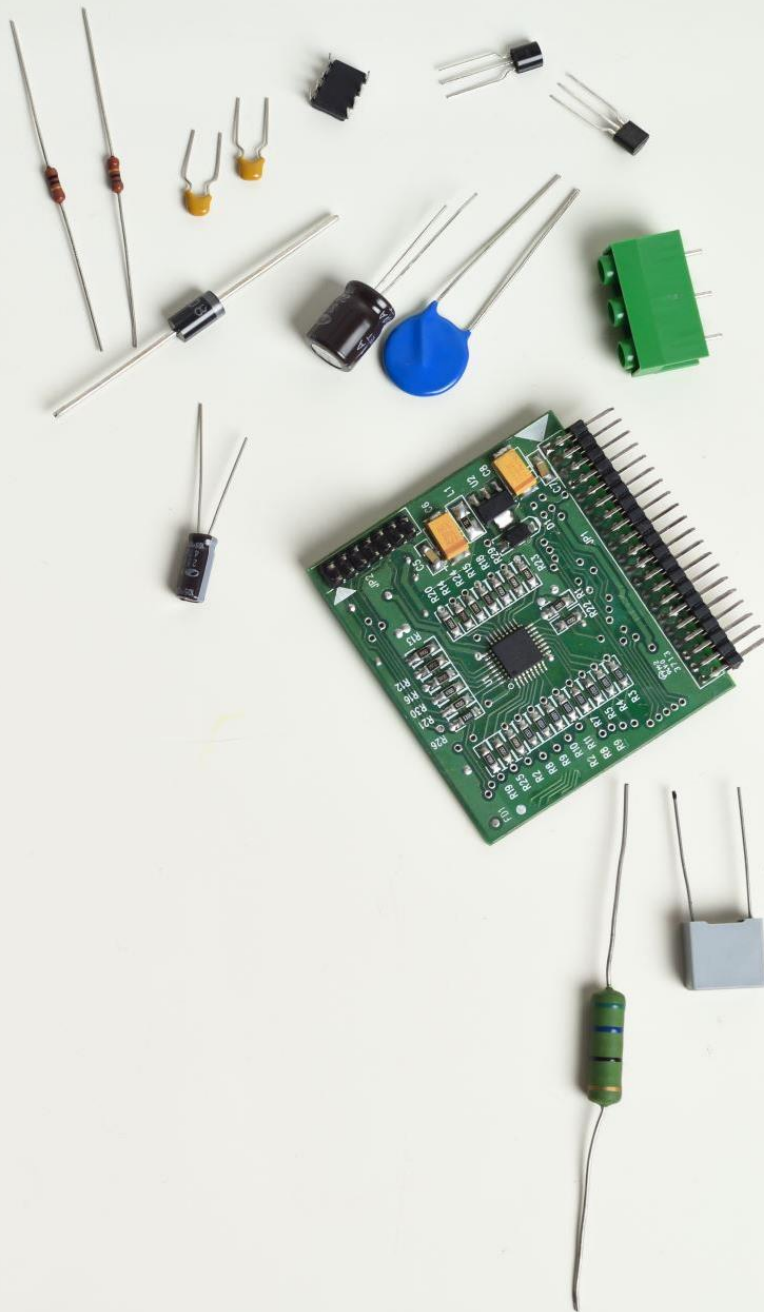

Industrial Electronics

BECS 44453 / ELEC 44163

Muditha Prasanna

Troubleshooting Basics

- Introduction to troubleshooting techniques and tools used in industrial electronic systems.



Common Components in Industrial Electronic Systems

- Common components such as sensors, actuators, controllers, and communication modules.

Types of Sensors

- Temperature Sensors: Thermocouples, RTDs
- Pressure Sensors: Strain gauges, piezoelectric sensors
- Proximity Sensors: Inductive, capacitive, ultrasonic
- Light Sensors: Photoresistors, photodiodes
- Flow Sensors: Turbine flowmeters, ultrasonic flow sensors
- Level Sensors: Capacitive, ultrasonic, float switches

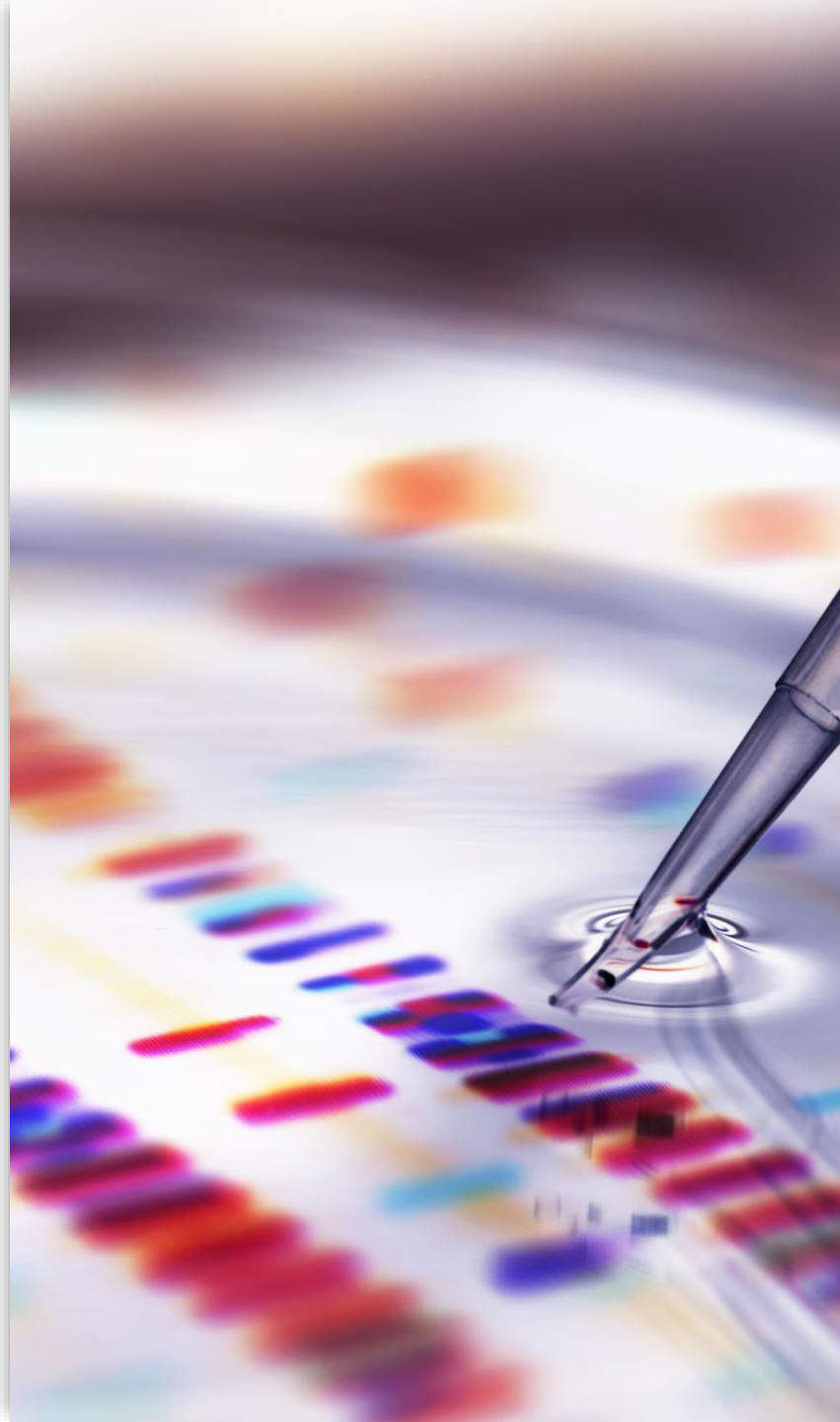
Controllers

Controller Functions

- Signal processing
- Decision making
- Output control
- Data logging and communication

Identifying Faults

Methods to identify faults in electronic systems, including visual inspection, signal tracing, and diagnostic software.



Visual inspection

Objectives of Visual Inspection

- Detect manufacturing defects
- Identify component damage or misplacement
- Ensure compliance with standards
- Prevent downstream failures

Common Defects Detected

Soldering issues (cold joints, bridges)

Misaligned components

Damaged components (cracks, burns)

Missing parts

Surface contamination or corrosion

Manual/ Visual Inspection

Manual /Visual
Inspection

Performed by trained
inspectors

Advantages: Flexibility,
human judgment

Limitations: Subjectivity,
fatigue, time-consuming

Automated Visual Inspection (AVI)

Use of cameras and image processing algorithms

Advantages: Speed, consistency, high precision

Technologies involved: Machine vision systems, AI, and pattern recognition

Equipment Used in Visual Inspection



Magnifying glasses
and microscopes



Automated Optical
Inspection (AOI)
systems

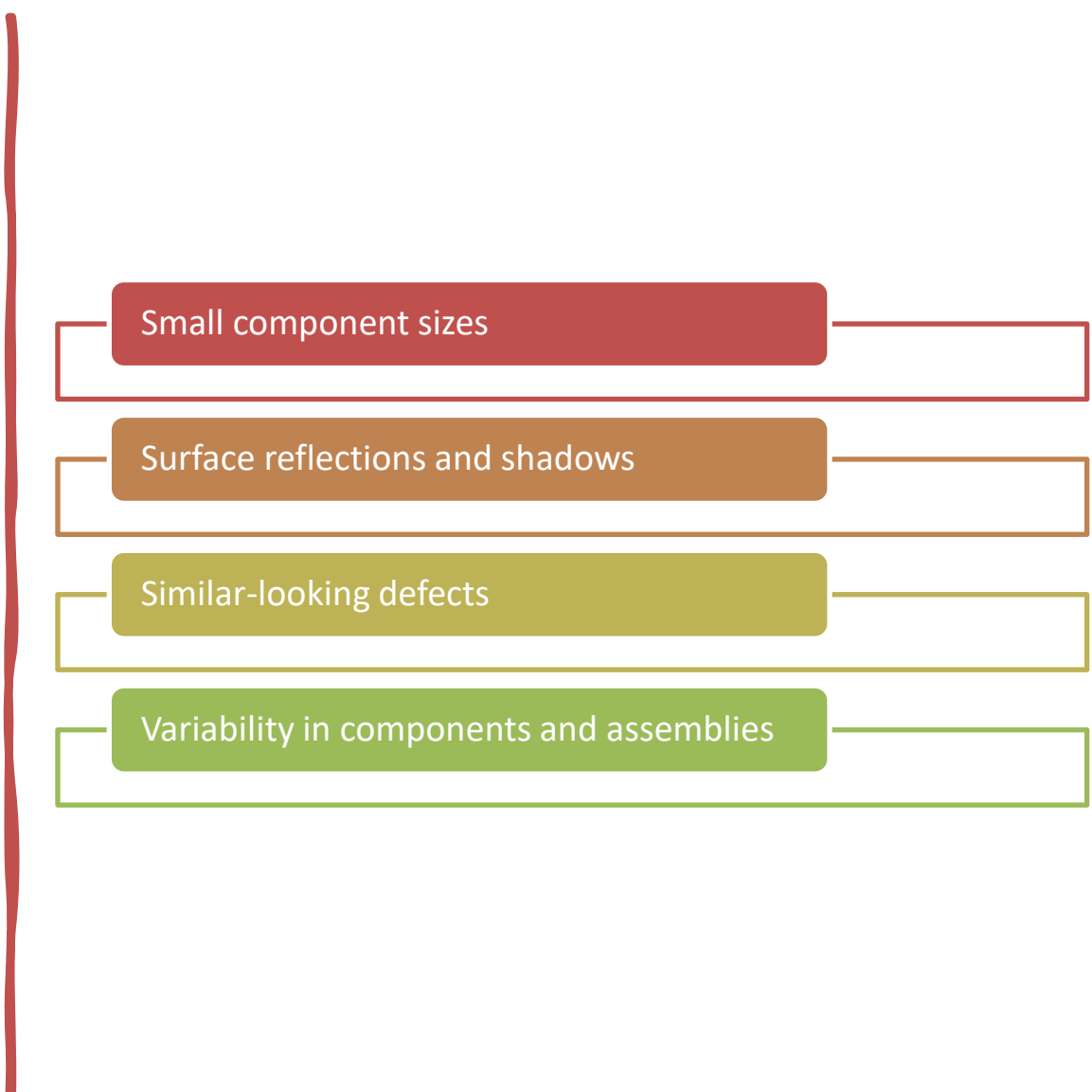


Cameras and
lighting setups



Software for image
analysis

Challenges in Visual Inspection



Small component sizes

Surface reflections and shadows

Similar-looking defects

Variability in components and assemblies

Best Practices

- Proper lighting and camera setup
- Regular calibration of inspection equipment
- Training inspectors thoroughly
- Combining manual and automated methods

Future Trends

- Integration of AI and machine learning
- 3D inspection techniques
- Real-time inspection and feedback
- IoT-enabled inspection systems

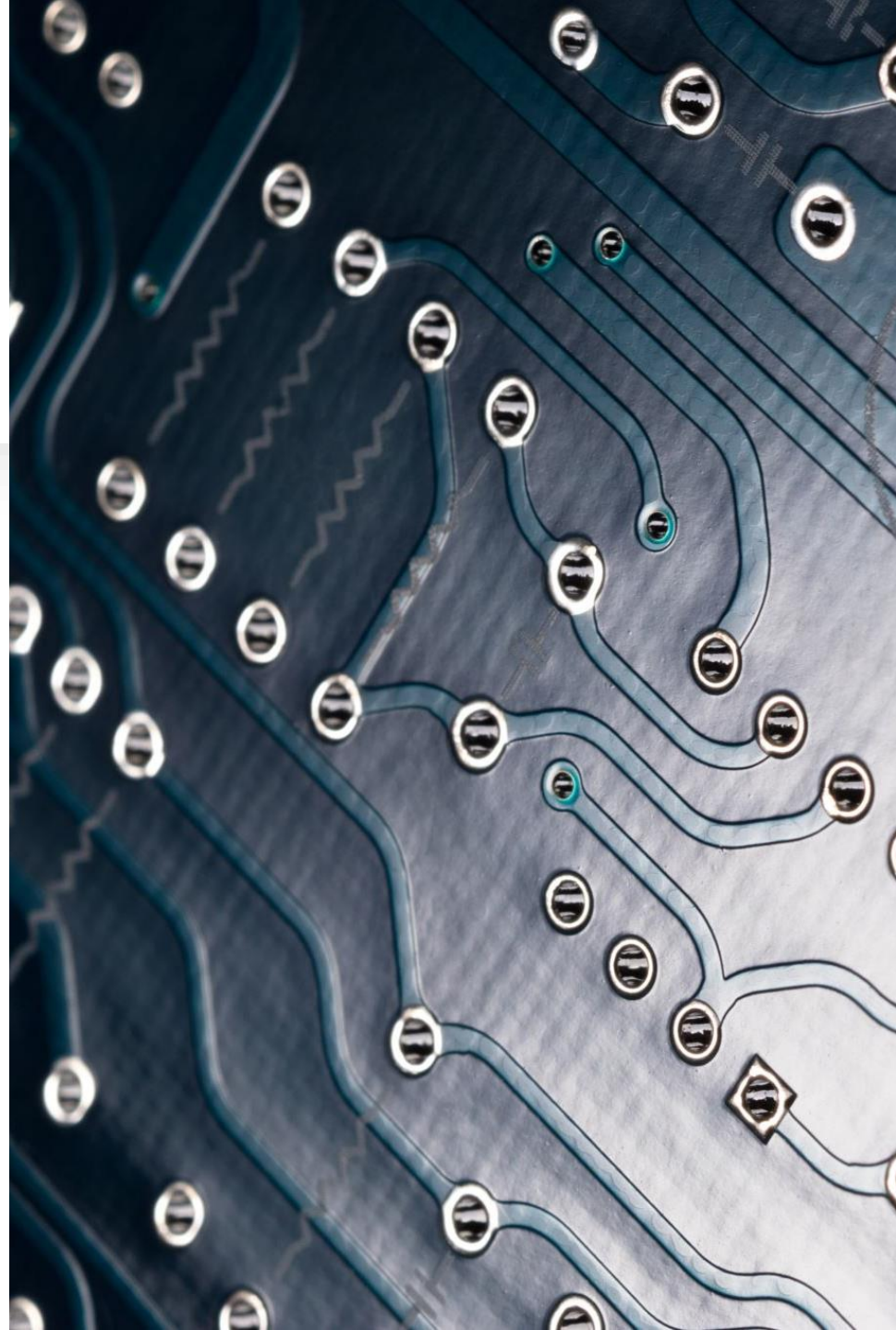
Signal tracing

What is Signal Tracing?

- Definition: Following the path of an electrical signal through a circuit
- Purpose: To locate faults or verify proper operation
- Used in diagnosing intermittent and hard-to-find issues

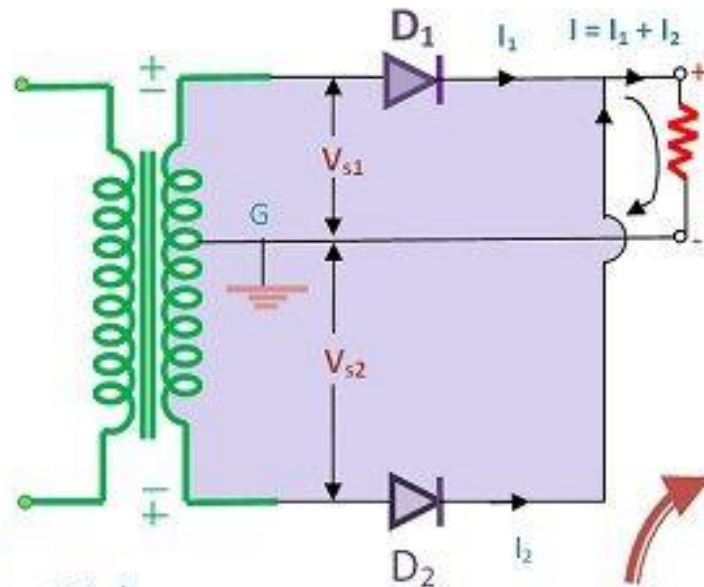
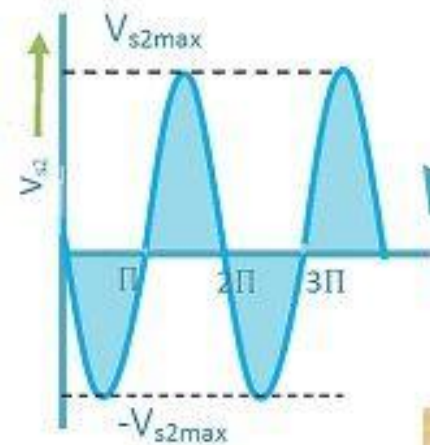
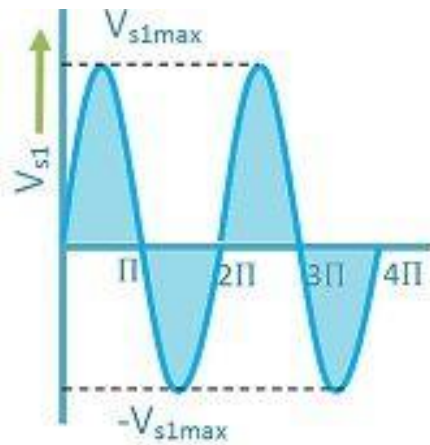
Why Signal Tracing is Important

- Identifies faulty components or connections
- Saves time compared to random testing
- Ensures circuit functionality
- Critical in complex electronic systems



Basic Principles of Signal Tracing

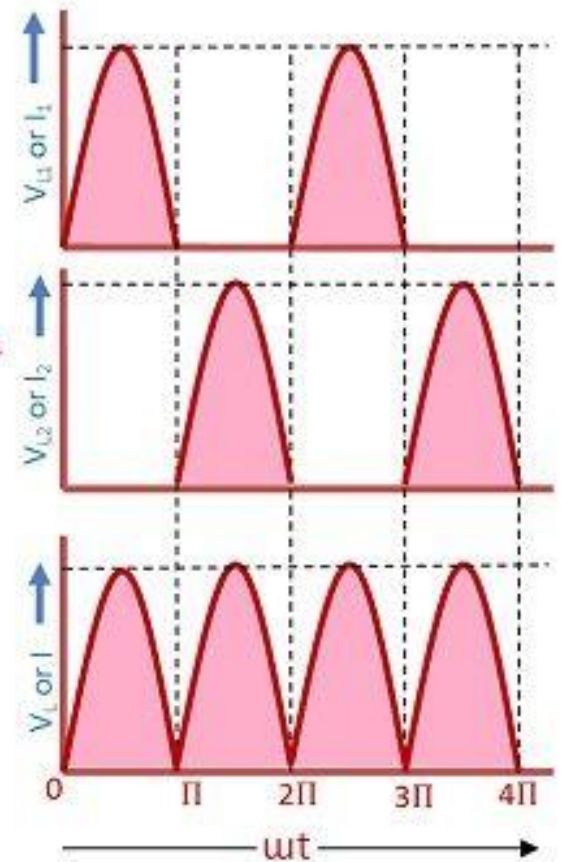
- Understanding circuit diagrams and schematics
- Using test equipment to observe signals at various points
- Comparing actual signals with expected waveforms



Input Voltage waveform

Output Voltage Waveform

Centre-tap full Wave Rectifier



Tools for Signal Tracing

- Multimeter (voltage, continuity)
- Oscilloscope (waveform analysis)
- Signal generator (if needed)
- Test probes and leads
- Test points in circuit design

Signal Tracing Techniques

- Visual inspection to identify obvious faults
- Powering the circuit and observing signals
- Using an oscilloscope to view waveforms
- Following the signal path step-by-step
- Comparing signals at different points

Common Signal Tracing Methods

- **Progressive method:** Moving sequentially through the circuit
- **Split method:** Dividing the circuit into sections
- **Branch method:** Testing different branches independently

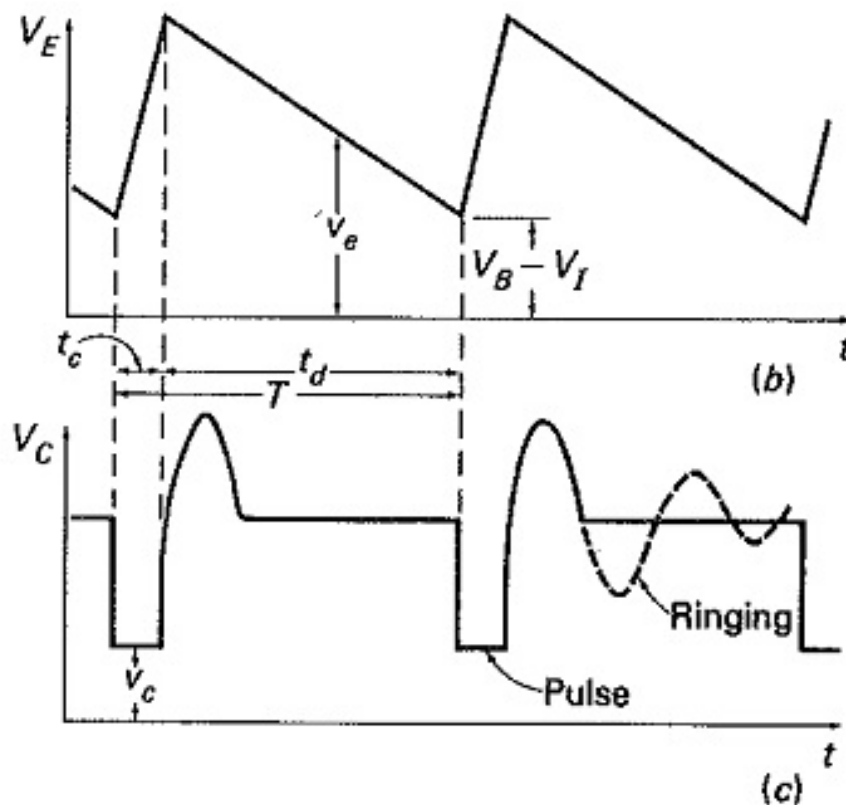
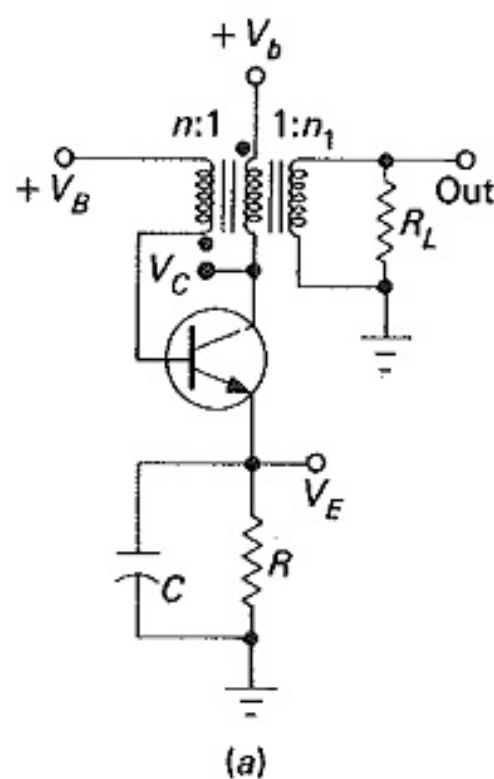
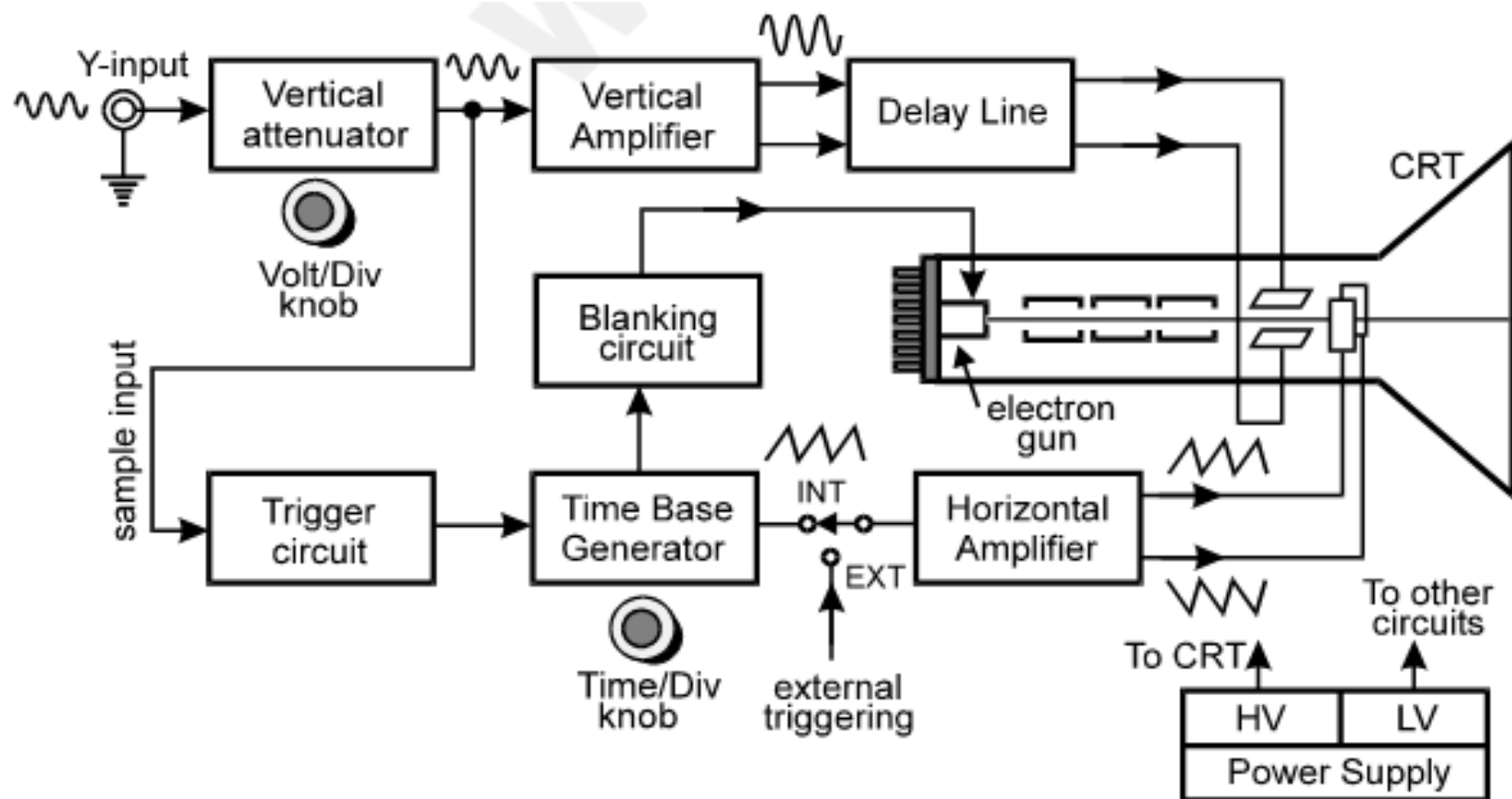


FIGURE 17-17 Blocking oscillator. (a) Basic circuit; (b) emitter waveform; (c) collector waveform.



Step-by-Step Signal Tracing Process

1. Understand the circuit and expected signals
2. Power on the circuit
3. Measure signals at key points
4. Identify where signals deviate from normal
5. Isolate the faulty component or section

Troubleshooting Tips

- Check power supplies first
- Confirm circuit ground connections
- Use reference waveforms for comparison
- Record measurements for analysis

Challenges in Signal Tracing

- Complex or dense circuits
- Intermittent faults
- Signal interference
- Small or hidden components

Best Practices

- Always refer to schematics
- Use proper safety precautions
- Start from the output and work backward or vice versa
- Document findings at each step

Diagnostic software

Software tools used to identify, analyze, and troubleshoot faults

Importance in modern electronics and industrial systems

Benefits: Faster diagnosis, accuracy, reduced downtime

What is Diagnostic Software?

- Specialized programs that interface with hardware or systems
- Collect data, run tests, and analyze system health
- Can be integrated into control systems or used as standalone tools

Types of Diagnostic Software

- **Embedded Diagnostic Software:** Built into devices for self-testing
- **PC-based Diagnostic Tools:** Run on computers to analyze embedded systems
- **Network Diagnostic Tools:** Monitor and troubleshoot networked systems
- **Industrial Control Software:** For PLCs, SCADA, and automation systems

Features of Diagnostic Software

- Real-time monitoring
- Fault detection and localization
- Data logging and reporting
- Graphical user interfaces (GUIs)
- Automated testing routines

Common Applications

- Troubleshooting electronic circuits
- Industrial automation system maintenance
- Network and communication systems
- Medical devices
- Automotive diagnostics

Benefits of Using Diagnostic Software

- Faster fault detection
- Precise identification of issues
- Reduced reliance on manual testing
- Historical data analysis for trend detection
- Predictive maintenance capabilities

Examples of Diagnostic Software Tools

- Oscilloscope and multimeter software
- PLC diagnostic tools
 - (e.g., RSLogix, Siemens STEP 7)
- Network analyzers
 - (e.g., Wireshark)
- Medical device diagnostic programs
- Automotive diagnostic scanners
 - (e.g., OBD-II tools)

PLC diagnostic tools - Siemens STEP 7

The screenshot displays the Siemens STEP 7 Ladder Addressing (LAD) editor interface. The main workspace shows two networks:

- Network 2: Timer**
A comment box states: "ODT timer provide on delay time i.e Timer will on After 10s. Rreset the timer ."
The ladder logic for Network 2 shows a normally open contact labeled **I0.2** connected to the **S_ODT** (Set On-Delay Timer) coil of timer **T1**. The timer's preset value **TV** is set to **S5T#10S**. The output of the timer is connected to coil **Q0.1**.
- Network 3: Pulse Timer**
A comment box states: "Pulse timer provide on delay time i.e Timer will on After 10s. Rreset the timer ."
The ladder logic for Network 3 shows a normally open contact labeled **I0.4** connected to the **S_PULSE** (Set Pulse) coil of timer **T2**. The timer's preset value **TV** is set to **S5T#10S**. The output of the timer is connected to coil **Q0.2**.

The left sidebar contains a project tree with the following structure:

- New network
- Bit logic
- Comparator
- Converter
- Counter
- DB call
- Jumps
- Integer function
- Floating-point fct.
- Move
- Program control
- Shift/rotate
- Status bits
- Timers
 - S_PULSE
 - S_PDT
 - S_ODT
 - S_ODTS
 - S_OFFDT
- Word logic
- FB blocks
- FC blocks
- SPB blocks
- SFC blocks

The bottom status bar shows the following information:

- Press F1 to get Help.
- Abt < 5.2 Nv 3 Rd Chg
- start
- New Text Document...
- SIMATIC Manager...
- HW Config - [SIMATIC...
- ST-PLC/CPM - [Simnew]
- LAD/STL/FBD - [G06...
- 11:01 AM

Network analyzers - Wireshark

The image shows the Wireshark network analyzer interface. The title bar reads "test.pcap - Wireshark". The menu bar includes File, Edit, View, Go, Capture, Analyze, Statistics, and Help. The toolbar contains various icons for file operations, capture, and analysis. The filter bar shows "tcp" with a dropdown menu for "Expression...", "Clear", and "Apply".

The main display area shows a list of captured packets. The columns are No., Time, Source, Destination, Protocol, and Info. The packets are as follows:

No.	Time	Source	Destination	Protocol	Info
11	1.226156	192.168.0.2	192.168.0.1	TCP	3196 > http [SYN] Seq=0 Len=0 MSS
12	1.227282	192.168.0.1	192.168.0.2	TCP	http > 3196 [SYN, ACK] Seq=0 Ack=
13	1.227325	192.168.0.2	192.168.0.1	TCP	3196 > http [ACK] Seq=1 Ack=1 Win
14	1.227451	192.168.0.2	192.168.0.1	HTTP	SUBSCRIBE /upnp/service/Layer3For
15	1.229309	192.168.0.1	192.168.0.2	TCP	http > 3196 [ACK] Seq=1 Ack=256 W
16	1.232421	192.168.0.1	192.168.0.2	TCP	[TCP Window Update] http > 3196 [
17	1.248355	192.168.0.1	192.168.0.2	TCP	1025 > 5000 [SYN] Seq=0 Len=0 MSS
18	1.248391	192.168.0.2	192.168.0.1	TCP	5000 > 1025 [SYN, ACK] Seq=0 Ack=
19	1.250171	192.168.0.1	192.168.0.2	HTTP	HTTP/1.0 200 OK
20	1.250285	192.168.0.2	192.168.0.1	TCP	3196 > http [FIN, ACK] Seq=256 Ac
21	1.250810	192.168.0.1	192.168.0.2	TCP	http > 3196 [FIN, ACK] Seq=114 Ac
22	1.250842	192.168.0.2	192.168.0.1	TCP	3196 > http [ACK] Seq=257 Ack=115
23	1.251868	192.168.0.1	192.168.0.2	TCP	1025 > 5000 [ACK] Seq=1 Ack=1 Win
24	1.252826	192.168.0.1	192.168.0.2	TCP	http > 3196 [FIN, ACK] Seq=26611
25	1.253323	192.168.0.2	192.168.0.1	TCP	3197 > http [SYN] Seq=0 Len=0 MSS
26	1.254502	192.168.0.1	192.168.0.2	TCP	http > 3197 [SYN, ACK] Seq=0 Ack=
27	1.254532	192.168.0.2	192.168.0.1	TCP	3197 > http [ACK] Seq=1 Ack=1 Win

The packet list is filtered by "tcp". The packet details pane shows the selected packet (Frame 11) with the following details:

- Frame 11 (62 bytes on wire, 62 bytes captured)
- Ethernet II, Src: 192.168.0.2 (00:0b:5d:20:cd:02), Dst: Netgear_2d:75:9a (00:09:5b:2d:75:9a)
- Internet Protocol, Src: 192.168.0.2 (192.168.0.2), Dst: 192.168.0.1 (192.168.0.1)
- Transmission Control Protocol, Src Port: 3196 (3196), Dst Port: http (80), Seq: 0, Len: 0

The packet bytes pane shows the raw data in hexadecimal and ASCII:

```
0000  00 09 5b 2d 75 9a 00 0b 5d 20 cd 02 08 00 45 00  ..[-u... ] ....E.
0010  00 30 18 48 40 00 80 06 61 2c c0 a8 00 02 c0 a8  .O.Hø... a,.....
0020  00 01 0c 7c 00 50 3c 36 95 f8 00 00 00 00 70 02  ...|.P<6 .....p.
0030  fa f0 27 e0 00 00 02 04 05 b4 01 01 04 02      ...'..... .....
```

The status bar at the bottom shows "File: 'D:\test.pcap' 14 KB 00:00:02" and "P: 120 D: 103 M: 0 [Expert: Error]".

CAR DETECTOR

MS309 DIAGNOSTIC TOOL OBD2



Read Codes



Erase Codes



View Live Data



View
Freeze Frame



Engine Check



Turn off MIL



I/M Readiness



Vehicle
Information



O2 Sensor Test



Free Shipping
USA Stock



Workflow of Diagnostic Software

1. Data acquisition from the system
2. Automated testing procedures
3. Data analysis and fault identification
4. Generating reports and recommendations
5. Implementing corrective actions

Challenges and Limitations

- Compatibility with diverse hardware
- Need for specialized knowledge
- False positives/negatives
- Dependence on software updates

Best Practices

- Keep diagnostic software updated
- Use in conjunction with manual testing
- Train personnel in software operation
- Regularly back up data and logs
- Integrate with maintenance management systems